Ex. 8.1 (Tut 1)

Q33 $\displaystyle\int_{-1}^{0} \sqrt{\frac{1+y}{1-y}}\, dy \cdot \sqrt{\frac{1+y}{1+y}}$

Conjugate of $1-y$
$= 1+y$

$$= \int_{-1}^{0} \sqrt{\frac{(1+y)^2}{1-y^2}}\, dy$$

$$= \int_{-1}^{0} \frac{\sqrt{(1+y)^2}}{\sqrt{1-y^2}}\, dy$$

$$= \int_{-1}^{0} \frac{|1+y|}{\sqrt{1-y^2}}\, dy$$

$$= \int_{-1}^{0} \frac{1+y}{\sqrt{1-y^2}}\, dy$$

$\because -1 \leq y \leq 0 \Rightarrow 1+y \geq 0$

$\displaystyle\lim_{\varepsilon \to 0} \int_{-1+\varepsilon}^{-\varepsilon} \sim$?

$\sqrt{x^2} = |x|$

$$= \underbrace{\int_{-1}^{0} \frac{1}{\sqrt{1-y^2}}\, dy}_{\arcsin y} + \int_{-1}^{0} \frac{y}{\sqrt{1-y^2}}\, dy$$

$u = 1-y^2$

$du = -2\boxed{y\, dy}$

Q32 $\int_{-1}^{1} \sqrt{1+x^2}\sin x \, dx = 0.$

Let $f(x) = \sqrt{1+x^2}\sin x$

odd func. $f(-x) = -f(x) \quad \forall x.$

$$f(-x) = \sqrt{1+(-x)^2}\,\sin(-x)$$

$$= \sqrt{1+x^2}\,(-\sin x)$$

$$= -\sqrt{1+x^2}\sin x$$

$$= -f(x).$$

$\therefore f$ is an odd function.

$\therefore \int_{-1}^{1} \sqrt{1+x^2}\sin x \, dx = 0.$

Q42

$$\int \frac{2^x - 1}{3^x}\,dx = \int \frac{2^x}{3^x}\,dx - \int \frac{1}{3^x}\,dx$$

$$= \int \left(\frac{2}{3}\right)^x dx - \int 3^{-x}\,dx$$

$$= \frac{\left(\frac{2}{3}\right)^x}{\ln\left(\frac{2}{3}\right)} - \left(-\frac{3^{-x}}{\ln 3}\right) + C$$

$$= \frac{\left(\frac{2}{3}\right)^x}{\ln\left(\frac{2}{3}\right)} + \frac{3^{-x}}{\ln 3} + C$$

$y = e^{\ln y}$

$f(x)^x$

$= e^{\ln f(x)^x}$

$= e^{x \ln f(x)}$

2^x

$\downarrow$

$e^{\ln 2^x}$

$e^{x \ln 2}$

Q18 $\int \frac{2^{\sqrt{y}}}{2\sqrt{y}}\,dy$

$(\sqrt{y})' = \frac{1}{2\sqrt{y}}$

$u = \sqrt{y}$

$du = \frac{1}{2\sqrt{y}}\,dy$

$= \int 2^u\,du$

$= \frac{2^u}{\ln 2} + C$

$= \frac{2^{\sqrt{y}}}{\ln 2} + C$

Q34 $\int e^{z+e^z}\,dz = \int \boxed{e^z} \cdot e^{e^z}\,\boxed{dz}$

$\boxed{u = e^z}$

$du = e^z\,dz$

$= \int e^{e^z}\,du$

$= \int e^u\,du = e^u + C = e^{e^z} + C.$

Q38

$$\int \frac{d\theta}{\cos\theta - 1} \cdot \frac{\cos\theta + 1}{\cos\theta + 1}$$

$$= \int \frac{\cos\theta + 1}{\cos^2\theta - 1}\, d\theta$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\cos^2\theta - 1 = -\sin^2\theta$$

$$= \int \frac{\cos\theta + 1}{-\sin^2\theta}\, d\theta$$

$$= -\int \frac{\cos\theta}{\sin^2\theta}\, d\theta - \int \frac{1}{\sin^2\theta}\, d\theta$$

① $u = \sin\theta$

$du = \cos\theta\, d\theta$

② $\frac{\cos\theta}{\sin\theta} \cdot \frac{1}{\sin\theta}$

"co"-family $= \cot\theta \csc\theta$

int $\longrightarrow -\csc\theta$

$\frac{1}{\sin^2\theta} = \csc^2\theta$

$\downarrow$ int

$-\cot\theta$

$$\int \frac{1}{1 - \csc\theta}\, d\theta \cdot \frac{1 + \csc\theta}{1 + \csc\theta}$$

Q39 $\displaystyle\int \frac{dx}{1+e^x} = \int \frac{1+e^x-e^x}{1+e^x}\,dx$

Numerator $= 1$

$= \underbrace{1+e^x}_{\text{in the denominator}} - e^x$

Hint: Long division !

$$= \int \frac{\cancel{1+e^x}}{\cancel{1+e^x}}\,dx - \int \frac{e^x}{1+e^x}\,dx$$

$u = e^x + 1 \Rightarrow du = e^x\,dx$

$$= x - \int \frac{e^x}{1+e^x}\,dx$$

$$\int \frac{e^x+1-1}{1+e^x}\,dx$$

$$= \int 1 - \frac{1}{1+e^x}\,dx$$

$$= x - \int \frac{1}{u}\,du$$

$$= x - \ln|u| + C$$

$$= x - \ln|\underbrace{e^x+1}_{>0}| + C$$

$$= x - \ln(e^x+1) + C$$

Q41

$$\int \frac{e^{3x} = e^{2x}\, e^{x}}{e^{x}+1}\, dx$$

$$\boxed{u = e^{x}+1}$$

$$\boxed{\frac{u^3}{u+1}}$$

$$e^{2x} = (e^{x})^2 \downarrow (u-1)^2 = u^2-2u+1$$

$$u = e^{x}$$

$$du = e^{x}\, dx$$

$$= \int \frac{u^2}{u+1}\, du$$

$$= \int \frac{\boxed{u^2-1}+\boxed{1}}{u+1}\, du$$

$$= \int \frac{u^2-1}{u+1}\, du + \int \frac{1}{u+1}\, du$$

$u^2-1 = (u+1)(u-1)$ in the denominator

$$= \int u-1\, du + \ln|u+1|$$

$$= \frac{u^2}{2} - u + \ln|u+1| + C$$

$$= \frac{e^{2x}}{2} - e^{x} + \ln(e^{x}+1) + C \qquad e^{x}+1 > 0$$